Citation Verification for AI Generated References

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***Abstract*—Through its GPT models OpenAI demonstrates significant capability to aid scholarly writing through automated tasks of citation generation which Artificial Intelligence assists with. The sophisticated language generation features of these models lead to citation hallucination where they produce realistic citations which point to nonexistent or inaccurate references. The academic research integrity faces substantial threats because of this situation.**

The main goal of this research involves performing a systematic evaluation of AI citations versus real Google Scholar references utilized as benchmarks. The project relies on SequenceMatcher from difflib within Python to determine how similar real and AI-text citations are when written in APA, MLA, Chicago, Harvard and Vancouver styles. The project analyzes patterns of inconsistent results by applying exploratory data analysis (EDA) with statistical hypothesis tests and data visualization of similarity distributions to assess AI-generated citation accuracy. The analysis exposes an urgent requirement for verification systems of AI-generated citations because the generated references frequently duplicate real citations but also show inaccuracies outside the level of acceptability. The research provides essential findings that help scientists, educational professionals and developers build better trust systems in AI academic output production

I. Introduction

Artificial intelligence systems have entered academic operations during recent years to create automated tools which help scholars handle literature searches and produce summaries along with generating citations more efficiently. OpenAI's GPT series along with other Large Language Models lead the most notable advances by producing academic documents with proper structure within seconds. The modern technological advances generate vast opportunities to improve research writing efficiency and accessibility but introduce the major problem known as citation hallucination.

The AI-generated issue of citation hallucination produces imaginary references which lack reality while also producing errors related to authorship attribution as well as title misidentification and incorrect publication year. Errors in academic citations compromise scholarship credibility because readers need genuine references to validate sources for future research development. Hallucinated citations which are not noticed will spread inaccurate information throughout the academic research community.

The academic sector presently lacks modern solutions to check the authentic nature of AI-generated citations despite increased AI tool usage in research. AI-generated model outputs pose a severe danger to novice academic researchers and students because they do not always recognize incorrectly generated content as inaccurate.

Given these circumstances this research addresses the fundamental question about AI-generated citation reliability and their match to real references. We built a systematic processing pipeline which retrieves real citations from Google Scholar and produces AI-produced citations for matching article titles before data normalization, data fusion and advanced string comparison. The research project aims to assess the reliability of AI-generated citations by generating actionable information dedicated to strengthening academic integrity during the AI era.

II. Review of Literature

Citation Hallucination in AI Systems

Researchers from [3] presented the first substantial evaluation of academic references hallucinations by showing how ChatGPT creates deceptive references which look real but reference non-existent publications. Academic research contains a threat to integrity through citation hallucination which introduces unverifiable false sources into the scholarly record. The research establishes the importance of establishing reliable verification systems which is the fundamental objective of this project according to [3]. Researchers in [7] studied whether ChatGPT-generated citations could be verified after testing its ability to identify retracted literature. Their data revealed that current systems failed to identify incorrect citations, demonstrating how academic content generated by AI without validation layers would deceive beginning researchers and students. The goal of our project matches the design of a validation pipeline as described in [7].

Cross-disciplinary Evaluation of Citation Accuracy
The authors in [12] evaluated large language models' citation generation performance across different academic disciplines and citation styles. Their cross-disciplinary framework revealed significant variation in citation accuracy, particularly in fields with lower representation in training data. Their methodology is highly relevant to this project as it highlights how discipline-specific citation verification may be necessary [12].

Taxonomies of AI Hallucination
The work in [8] introduces a complete catalogue of hallucinations in natural language generation that reveals citation errors as extrinsic hallucinations generated text referencing unreal facts or unreal sources. The survey examines hallucination detection techniques involving uncertainty modelling and retrieval-augmented generation that apply directly to our system's citation verifiability enhancement [8].

Cognitive Limits of LLMs in Citation Generation
The authors in [4] introduce mathematical interpretations combined with an explainability framework to assess transformer-based language model processing of structured tasks. Their research demonstrates LLMs produce fluent text output yet they rely on pattern recognition between tokens instead of building true symbolic or relational comprehension. Structure preservation in output tasks becomes essential for assignments like citation generation because relationships among metadata elements (such as author and title) need maximal semantic maintenance. Experience from our research effort supports the implementation of structured comparison tools including reference similarity scoring between samples [4].

Citation Errors in Summarization and Generation Tasks
A detailed analysis of ChatGPT across different languages and tasks is presented in [1] which focuses on detecting hallucinations in both reasoning steps and information retrieval outputs. Factual inquiries from ChatGPT frequently result in believable yet inaccurate responses which incorporate both fake lists and non-existent citations according to the study. Our research confirms citation hallucination findings which support the necessity of using similarity matching and metadata normalization tools for maintaining academic content citation fidelity [1].

Retrieval-Augmented Generation Approaches
The hybrid model in [9] blended retrieval fundamentals with generation methods to enhance response factual accuracy. The proposed method achieved substantial hallucination reduction because it linked outputs to retrieved documents. Success indicators from this model indicate that upcoming citation tool versions must include retrieval-based elements which could serve as an enhancement plan for our present system framework [9].

Verifiability in AI Search Outputs
Researchers in [11] established a verification protocol to determine if search engine outputs produced by AI and including citation references can be verified through public databases. The research discovered numerous AI-generated citations failed validation standards which supports the requirement for formal verification processes like the project presented here [11].

Classification of Factual Errors in LLMs
Islam et al. in [13] explore an in-depth overview of the thirty-two different methods that assist with reducing hallucinations in large language models (LLMs). The paper presents a detailed classification system for these methods which organizes them according to different criteria including dataset usage patterns and common tasks and feedback mechanisms and retriever types. The classification scheme helps researchers distinguish unique approaches designed for mitigating hallucinations in LLMs. The research supports our project goal to develop detection and mitigation methods for citation inconsistencies in AI-generated content [13].

Ethical and Practical Challenges of Large Language Models
The author [2] engages with large language models from an expanded viewpoint by demonstrating how productive outputs tend to conceal factual errors. The authors outline dangerous risks linked to trusting unverified outputs from programs which remain difficult to audit particularly when used in academic research. The research in [2] supports our ethical drive to develop a transparent statistical framework for citation verification [2].

Entropy-Based Detection of Hallucinated Citations
Farquhar et al. present in [5] a new technique to detect hallucinations in Large Language Models (LLMs) which uses semantic entropy evaluation. The approach quantifies model-generated output uncertainty to identify cases where models demonstrate excessive confidence in information that turns out to be incorrect or fabricated. This detection method shows special relevance to hallucinated citations since it establishes numerical criteria for identifying unreliable LLM-generated references. This research establishes the necessity of uncertainty estimation methods in LLM outputs to improve the trustworthiness of AI-generated academic materials [5].

Inference-Time Techniques for Enhancing Truthfulness in LLMs
Li et al. present Inference-Time Intervention (ITI) as an intervention technique in [10] which optimizes large language model (LLM) truthfulness through activation modifications during inference time. The performance of LLaMA models on TruthfulQA benchmark was notably enhanced through ITI which allows researchers to control the activations across focused attention heads. The application of Inference-Time Intervention (ITI) techniques on an instruction-finetuned LLaMA model named Alpaca led to a 32.5% increase in truthfulness reaching 65.1%. The method requires a small number of samples to find truthful directions while remaining lightweight and noninvasive to computational processes. Evidence shows that LLMs operate with an internal appreciation of truth likelihood while generating textual responses containing false content. The discoveries in this research directly contribute to our project because they demonstrate that online interventions performed during inference operations can lower the frequency of false citations in AI-created academic manuscripts [10].

Factored Verification for Hallucination Detection in

Academic Summarization

The method proposed in [6] is called Factored Verification, and it is an automatic method aimed at identifying hallucinations in the abstractive summaries of academic papers. This approach breaks down summaries to claims and then evaluates the accuracy of the claims against the documents using language models. Their method also presented high accuracy of seventy-six percent two on the HaluEval benchmark which made it achieve what they termed as new state-of- the-art. Specifically, for summaries produced by the models such as ChatGPT, GPT-4, and Claude 2, it was possible to detect an average of 0.62 – 1.55 hallucinations per summary. Introducing model self-correction based on these verifications led to a 30% reduction in hallucinations. This paper emphasizes the need to incorporate verification techniques within LLM outputs, in line with our project of checking the facts and accuracy of the citations produced by AI [6].

Multilingual Benchmarks for Hallucination Detection in LLMs

Poly-FEVER published by Zhang et al. [14] is a multilingual corpus for evaluating LLM's strengths in identifying hallucinations. The dataset comprises of 77,973 factual assertions with their corresponding labels, collected from datasets such as FEVER, Climate- FEVER and SciFact. This resource helps in a more standardised manner, evaluate LLMs' ability in perceiving hallucinations in different language environments. In this respect, it is worth mentioning that elucidated factors include the distribution of topics and available web linkages that point to the impact of language biases in the models. Through the proposed cross-linguistic evaluation, Poly-FEVER aids in the development of more accurate and non-English-focused AI systems, which aligns with the purpose of our proposal: vide ensuring factual accuracy of citations generated by AI [14].

III. Objectives of the Study

The main goal of this research involves creating a system which measures AI-generated citation reliability through the evaluation of references from Google Scholar. The project targets the trust deficiency between AI output validation and academic requirements through an objective numeric assessment system.

The project follows these particular objectives:

- The first objective includes data collection through metadata scraping from Google Scholar with the goal of achieving broad academic discipline representation.
- The research will produce AI references from OpenAI's GPT-3.5-turbo API which generates five forms of citations including APA, MLA, Chicago Harvard and Vancouver.
- The datasets require normalization through lowercasing while eliminating whitespace followed by converting field formats to a standard format which allows for accurate assessment.
- The normalization process of both Title and Author fields allows merging datasets so original citations and AI-created ones align correctly for the same publications.
- The textual similarity scores between real and AI citations will be calculated through Python's difflib SequenceMatcher while determining actual matching levels.
- We utilized one-sample t-tests for statistical hypothesis testing to verify if the observed similarity matches the perfect score of 1.0.
- The analysis combines histograms and descriptive statistics to show the similarity distribution patterns which help interpret the data patterns.
- This investigation aims to determine the dependability levels along with safety concerns about dependency on AI-generated citations which lack manual verification.

The project's success aids the development of AI-based academic writing tools with increased confidentiality and validate scholarly communication protection.

IV. Methodology

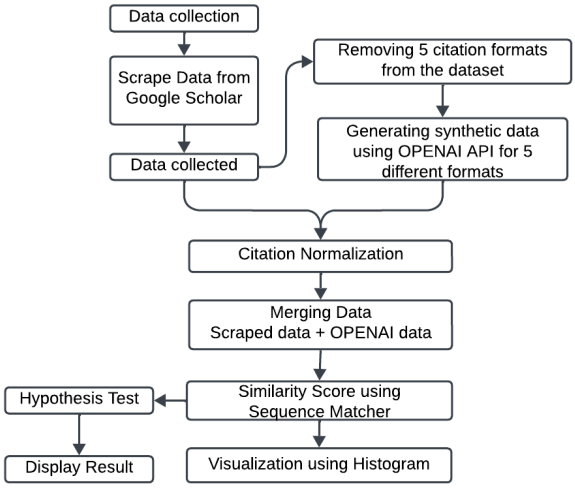

Fig. 1. Research Methodology

V. Data Collection

Any analytics project starts with data collection as its necessary initial step. Our study needed two distinct data sets which included original citations as benchmarks together with AI-generated citations for evaluation.

Research data was obtained through a Google Scholar scraping operation from this academic database that serves scholarly publications across different subject areas. Through the implementation of Python's Selenium library an automatic Google Scholar search system was built to retrieve essential metadata such as Title, Author(s), Document Type and Year of Publication along with Source Link (URL) data. The database Google Scholar was selected for its trustworthy operations and wide content range and open access to citation information.

We obtained the second dataset through an API interface with GPT-3.5-turbo from OpenAI. The Google Scholar document titles triggered OpenAI to produce formal citations that conformed to APA 7th Edition and MLA 9th Edition

and also to Chicago 17th Edition and Harvard along with Vancouver. The specification prompt system guided the model to create formatted citations which strictly followed the established standards of each citation style. The system delivered consistent results throughout outputs and controlled the initial design modifications that could affect formatting.

The datasets used structured file formats that included Excel spreadsheets and CSV files to provide easy access for processing needs. AI-produced citations matched the original real citations through shared document titles and authors for the purpose of meaningful later comparison. The use of two data sources enabled the creation of an environment that supported controlled tests for verifying AI citation accuracy.

VI. Exploratory Data Analysis (EDA)

Exploratory data analysis (EDA) assessed the obtained datasets following data collection so analysts could prepare raw evidence for formal examination.

The data quality system performed its initial checks at this stage. The essential variables Title Author(s) Document Type and Citation appeared with complete entries in every observation. Critical field values containing missing or blank entries either received corrections (when possible) or the rows received exclusion to keep the data sets intact.

The normalization process was executed to both data collections. All names of titles became lowercase for data normalization purposes to avoid inequality between capitalization systems. A cleanup process eliminated initial and final spaces so the systems would avoid merging errors. Standardization of formatting involved cleaning up excessive word spacing found in text fields.

An initial display of the document distribution between journal articles and conference papers demonstrated the publication year range. The analysis confirmed that the data collection process included various publication types without bias toward any type.

We developed official statements from our EDA findings which would direct the statistical analysis: The AI-generated citation shows no meaningful differences from genuine citations according to this null hypothesis which states that the mean similarity score equals 1 (μ=1). The mean similarity score (μ) equals 1.

$$H_0 : \mu = 1$$

The experimental data indicate a substantial dissimilarity between the artificially generated citations and real instances through the alternative hypothesis (H_1). The calculated mean similarity measure falls below one.

$$H_1 : \mu < 1$$

These hypotheses were essential because statistical testing would determine if the differences observed were actual or random in nature. Our testing of hypotheses sought to officially determine if artificial intelligence citations displayed lesser accuracy levels than those from real sources. One-sample t-test formula:

$$t = \frac{\bar{x} - \mu_0}{s/\sqrt{n}} \tag{1}$$

Where:

- $\bar{x}$ = sample mean
- μ_0 = hypothesized mean (1.0)
- s = sample standard deviation
- n = sample size

VII. Data Analysis

Throughout the data analytics phase the project executed its main technical process through several interconnected steps.

The two citation datasets which originated from Google Scholar and OpenAI were joined together following normalization procedures. That normalized Title and Author fields defined the basis for an inner join which let the AI citations match correctly to real citations. A unified dataset formed that incorporated complete information needed for direct comparison between rows.

After data merging operations came the assessment of citation style similarity between real and AI-generated citations. The Python library difflib features SequenceMatcher which we used as part of our method. SequenceMatcher analyzes sequences in text between two strings to calculate a matching ratio according to this formula:

$$\text{Similarity Ratio} = \frac{2.0 \times \text{Matching Characters}}{\text{Total Characters in Both Strings}} \tag{2}$$

We utilized SequenceMatcher to determine five similarity ratings for each document by analyzing the citation styles APA, MLA, Chicago, Harvard and Vancouver independently.

A statistical analysis evaluated AI-generated citations to determine their differentiation from authentic citations after similarity scores calculation. The study conducted a one-sample t-test as a statistical approach to confirm the hypotheses created during the EDA phase. The one-sample t-test compares the mean similarity score to the expected mean of 1 (perfect match).

VIII. Data Visualization and Results Report

The next operation focused on analyzing the citation similarity score distributions across styles to detect patterns and abnormal readings. The role of visualization stands essential for converting sophisticated numeric findings into simple visual insights since it helps users decipher the degree of AI citation similarity to genuine citations.

The citation styles APA, MLA, Chicago, Harvard, and Vancouver received separate histogram plots through Matplotlib in Python. The graphs presented frequency distributions depicting similarity score data points where x-axis numbers indicated score range from 0.5 to 1.0 and y-axis numbers represented the number of citations per range. The graphics displayed multiple crucial trends that became evident.

AI-generated citations produced scores between 0.75 to 0.95 with most citation styles which indicates imperfect accuracy in their generation.

The average similarity scores of APA-style citations appeared higher than others because the rigid structure of APA formatting seems best suited for AI model duplication. The research findings showed the citation styles failed to create clusters around the 1.0 mark marking the absence of perfect matches throughout the evaluation process.

The variations in Vancouver citations stemmed from inconsistent journal abbreviation conventions and the problem AI models face when creating medical citations accurately.

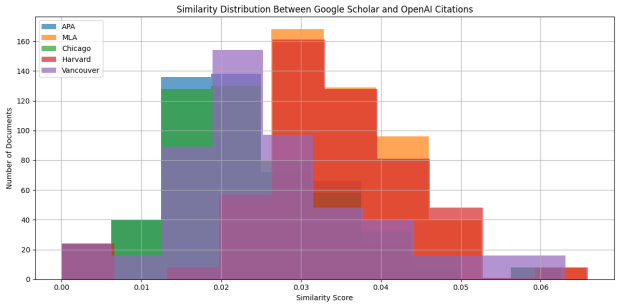

Fig. 2. Similarity Score Distribution

The histograms revealed outlier AI citations which scored below 0.7 on similarity checks the analysis showed contained major differences such as author omissions and mistakes or incomplete journal title information. The lower-scoring citations pointed out potential hallucinations together with severe citation errors.

- Mean similarity score
- Median score
- Standard deviation
- Min and Max scores

IX. Findings

- AI-generated citations delivered results with metadata and formatting accuracy between 0.75 and 0.95 while falling short of complete accuracy.
- APA-style citations achieved the highest similarity scores because their strict formatting guidelines match the replication capabilities of LLMs like GPT-3.5.
- The Vancouver-style citation system performed poorly in terms of consistency because of conflicting abbreviations and ambiguous formatting standards and name normalization issues.
- The samples with low similarity scores showed substantial errors which included made-up author names together with errors in publication dates and imaginary journal names.
- The one-sample t-tests across styles found mean similarity scores significantly lower than 1.0 which proved citation inaccuracies to exist in LLM-generated outputs.
- The detection of outlier citations using visualization and distribution analysis revealed structural hallucinations that highlight the need for automated citation verification before academic use.

X. Conclusion

The examination shows that OpenAI's GPT-3.5 model produces citations which demonstrate a major reliability issue. The study performs a thorough analysis of AI-generated versus authentic Google Scholar citations to identify regular shifts in citation formatting along with author identification inconsistencies and reference accuracy issues. Python's SequenceMatcher measured quantitative similarities between APA MLA Chicago Harvard and Vancouver styles which found that properly formatted outputs frequently contained factual errors.

The combined evidence from exploratory examination together with statistical tests and distributed visual assessment demonstrates current LLM citation generation suffers from a persistent structural problem in citation hallucination. Structurally valid AI-generated citations exist but their content does not match actual academic references verifiably. The detection of low-similarity outlier cases highlights the risks involved in using these tools for scholarly work when no human oversight exists.

Analysis of existing studies confirmed our experimental results since organizations throughout different fields encounter similar issues. Research findings including hallucination typologies and retrieval-based grounding methods combined with truthfulness-enhancing techniques such as Inference-Time Intervention together with entropy-based detection confirmed the necessity for solid post-generation verification systems.

The proposed pipeline delivers a verification framework using data-driven approaches which evaluate AI citation reliability by employing real citation benchmarks with normalization techniques and statistical testing methods. The research findings help expand AI-assisted academic writing knowledge by proposing to build integrated trust features into citations for future LLM infrastructure and educational research platforms.

The study intends to evaluate the academic quality of large language model outputs rather than dismiss the tools' role in scholarly work. The eventual integration of real-time verification protocols together with retrieval-based augmentation stands to significantly reduce hallucination risk which will pave the way for responsible academic writing through artificial intelligence systems.

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